Meta-Learning for Rapid Personalization of Error-Related Potential Brain-Computer Interfaces: A Two-Dataset Benchmark

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Abstract*—When a brain-computer interface (BCI) makes a mistake, the user's electroencephalogram (EEG) carries a telltale error-related potential (ErrP) that the system can use to detect and correct itself. The difficulty is that ErrPs look different from one person to the next, so a fresh user typically has to sit through a long calibration session before the decoder is any good. We ask whether meta-learning can shorten that session. Instead of fitting a separate model per user, a meta-model learns an initialization across users that a new user can adapt from a few class-balanced calibration trials. We compare ten approaches under one leave-one-subject-out (LOSO) protocol, run independently on two structurally different ErrP datasets (we do not train on one and test on the other): three gradient-based meta-learners (MAML, ANIL, Reptile), two metric-based ones (Prototypical and Matching networks), a subject-conditioned encoder built on feature-wise linear modulation (FiLM), and four classical baselines (a supervised MLP, EEGNet, and two Riemannian methods), plus two transfer controls that share the source-subject data. At a five-trial calibration budget, the gradient-based meta-learners, the FiLM encoder, and a source-pretrained EEGNet all significantly improve balanced accuracy over a from-scratch supervised baseline on both datasets (paired Wilcoxon at the subject level, $N{=}16$, BH-FDR $p < 0.05$). The advantage *replicates* on the second, independent paradigm rather than transferring to it: there, at $K{=}5$, the gradient meta-learners, the FiLM encoder, and Prototypical networks all outperform EEGNet by several points, though this edge closes by $K{=}10$. The meta-learners also surpass transfer baselines that pretrain on the same source subjects with the same class-weighted loss; because test-time adaptation still differs, we attribute the residual gain to the meta-learning *approach* (episodic training with gradient adaptation) rather than data access alone. Ablations show that class-weighted episodic loss is the most important design choice, while FiLM conditioning and feature dimensionality contribute little. We frame the result as improved low-data personalization, not a deployment-ready decoder.

***Index Terms*—brain-computer interface, error-related potential, meta-learning, few-shot learning, EEG, personalization**

I. Introduction

A brain-computer interface (BCI) turns EEG into commands, and like any decoder it sometimes gets them wrong. When it does, for instance when a P300 speller picks the wrong letter, the user's brain reacts with a stereotyped *error-related potential* (ErrP): a fronto-central deflection with an early negativity around 250 ms and a later positivity around 320 ms after the erroneous feedback [1], [2]. If the system can read that signal, it can undo the mistake on its own, with no extra effort from the user [3].

Calibration is what stands in the way. Because ErrP morphology shifts with anatomy, electrode placement, and signal quality, a decoder trained on one person transfers poorly to the next, and the usual remedy is to record tens of minutes of labelled trials from every new user before the system works. That per-user overhead is a large part of why ErrP-BCIs rarely leave the lab [4].

Meta-learning offers a way around it. Rather than learning one decoder per user, a meta-model trained across many users learns an initialization that adapts to a new user from a handful of trials [5]. We put this idea to the test and report three things:

- a head-to-head benchmark of ten personalization methods—three gradient meta-learners, two metric meta-learners, a subject-conditioned FiLM encoder, and four classical baselines—all evaluated under one leave-one-subject-out protocol;
- replication on a second, structurally different ErrP paradigm (run as a separate LOSO experiment, not trained across corpora), where the advantage holds and stays stable across three random seeds; and
- ablations that pin down which design choices actually carry the result.

We are not proposing a new algorithm. The value here is the benchmark, the two-dataset evidence, and the analysis of what makes few-shot ErrP decoding work.

II. Related Work

ErrP detection. ErrPs have been used both to correct BCI output and to supply reward signals to autonomous agents [1], [3]. Detecting them in a single trial is still hard. Recent surveys weigh linear discriminants, support-vector machines, and Riemannian classifiers against one another, and they keep returning to the same two practical issues: preprocessing choices and the heavy class imbalance of ErrP data [2], [4]. Convolutional decoders such as EEGNet and the deep/shallow ConvNets have become the standard neural baseline for EEG, including ErrP detection [6], [7]; we therefore treat a source-pretrained EEGNet as the reference our few-shot methods must beat.

Cross-subject transfer for EEG. Reducing per-user calibration is a long-standing goal. Beyond meta-learning, the dominant line is domain adaptation—most prominently Riemannian/covariance alignment and Euclidean alignment of trials before a shared classifier [8], [9], surveyed comprehensively by Wu et al. [10]. These methods adapt the feature distribution rather than learning to adapt; we include Riemannian and covariance-alignment baselines as their representatives, and a pretrained fine-tuning baseline as the transfer-learning analogue of our meta-learners.

Meta-learning. MAML learns an initialization that adapts in a few gradient steps [5]; ANIL shows that adapting only the classifier head usually suffices [11]; and Reptile reaches a similar place with first-order updates alone [12]. The metric-based alternatives, Prototypical [13] and Matching networks [14], instead classify by similarity in a learned embedding. Feature-wise linear modulation (FiLM) conditions a network on side information [15]; we use it to condition on a subject's support set.

Meta-learning for EEG. Most prior EEG work targets motor imagery: MAML and relatives for cross-subject decoding [16], [17], and meta-learned representations for zero-calibration use [18]. Few-shot EEG classification has grown into an active subfield [19]. Single-trial ErrP personalization has received far less attention, and we are not aware of prior work that validates it across datasets. That is the gap we address.

III. Methods

A. Datasets

We use two independent ErrP datasets. The **primary** dataset is the INRIA BCI Challenge (NER 2015) P300 speller (16 subjects, feedback-locked ErrP paradigm) [20]. The **validation** dataset is the ErrP-Coadaptation corpus [21]; we use all 16 subjects in the public release (indexed S03–S18) with no exclusions of our own. It is an observation-ErrP paradigm in which a participant infers a robot's intended object from its gaze. The two paradigms differ in task, montage, channel set, and sampling rate. We run the *same* LOSO protocol *separately* on each dataset; we do *not* train on one and test on the other, since the corpora are not channel- or rate-matched. "Validation" therefore denotes *replication* of the finding on a second, independent paradigm—evidence that the result is not idiosyncratic to one corpus—rather than out-of-distribution transfer between corpora.

B. Preprocessing

We applied a 4th-order zero-phase Butterworth bandpass (1–40 Hz) to the continuous EEG before epoching to avoid edge artifacts. The INRIA recordings are sampled at 200 Hz; the coadaptation recordings at 250 Hz, where a 50 Hz notch was applied before the bandpass to suppress line noise (for the 200 Hz INRIA data the 40 Hz low-pass already removes the 50 Hz line component, so no separate notch is used). No resampling was performed; each method's time-domain sizes follow the native rate. Epochs were cut from -200 to $+600$ ms relative to feedback onset; per-epoch per-channel baseline correction was applied over $[-200, 0]$ ms; epochs exceeding $\pm 100\,\mu$V peak-to-peak were rejected, and K refers to clean trials after rejection. Epoch–label alignment used the feedback index to prevent silent offset errors. The gradient and metric meta-learners operate on the flattened filtered waveform reduced to 32 whitened principal components (PCA fit on training subjects only within each fold to prevent leakage); EEGNet and the Riemannian methods operate on the raw filtered epochs. This input asymmetry follows each method family's standard usage but is a confound for cross-family comparison; the feature-representation ablation (Sec. V) addresses it by varying the PCA dimensionality, and "raw" there denotes the flattened waveform at a higher PCA rank rather than the time-domain tensor fed to EEGNet.

C. Personalization methods

We compare, under a K-shot protocol (the new user's calibration set is *class-balanced*, $K/2$ trials per class, so an error trial is always present even at $K{=}5$): **Full-MAML** [5], **MAML-ANIL** [11], and **Reptile** [12] (gradient-based); **Prototypical** [13] and **Matching** [14] networks (metric-based); and **SubjectConditioned**, an encoder that computes error/correct prototypes from the support set and modulates a feature extractor via FiLM [15]. All episodic losses are class-weighted (inverse class frequency) to handle the ∼75/25 correct/error imbalance.

D. Baselines

Supervised: a three-layer MLP (128-64-2 units, LayerNorm, GELU, Dropout 0.3) trained from scratch on K trials with early stopping, using a class-weighted loss. It is the only method that does *not* see source-subject data. **EEGNet** [6]: a compact depthwise-separable CNN whose backbone is pretrained on the $N{-}1$ source subjects with a class-weighted cross-entropy loss, then frozen; an ℓ_2-regularized logistic-regression probe (class-balanced, lbfgs) is fit on the K support embeddings. EEGNet is therefore itself a source-pretrained transfer baseline with the same data access as the meta-learners, not a from-scratch method. **Riemannian** [22] and **Covariance Alignment** [8], [23]: covariance-based classifiers on the SPD manifold; their xDAWN spatial filters and tangent-space projection are fit on the source subjects, and only the final LDA uses the K trials—so these too are source-aware. To further separate the meta-learning approach from data access, we add two *fair-transfer* baselines that see exactly the same $N{-}1$ source subjects: **Pretrain-FT** pretrains the same MLP backbone on all source subjects with the *same* class-weighted loss used by the meta-learners, freezes it, and fits a class-balanced logistic-regression probe on the K new-user trials; **Pretrain-ZeroShot** keeps that pretrained backbone and its source-trained head and applies no new-user adaptation at all. These baselines match the meta-learners on data access and

TABLE I
METHOD CHARACTERISTICS. SRC. = USES SOURCE-SUBJECT DATA; CW = CLASS-WEIGHTED LOSS; INPUT IS WHAT THE CLASSIFIER INGESTS (PCA = 32 WHITENED COMPONENTS OF THE FLATTENED WAVEFORM; FLAT = FLATTENED WAVEFORM; EPOCHS = RAW $C \times T$ TENSOR).

Method	Src.	CW	Test-time adaptation	Input
Supervised	no	yes	train from scratch (K)	PCA
Pretrain-ZeroShot	yes	yes	none	PCA
Pretrain-FT	yes	yes	LR probe (frozen)	PCA
Full-MAML	yes	yes	gradient, full (5 steps)	PCA
Reptile	yes	yes	gradient, full (5 steps)	PCA
MAML-ANIL	yes	yes	gradient, head (5 steps)	PCA
Subj.-Cond.	yes	yes	FiLM + gradient	PCA
Prototypical	yes	bal. ep.	nearest-prototype	flat
Matching	yes	bal. ep.	attention kNN	flat
EEGNet	yes	yes	LR probe (frozen)	epochs
Riemannian	yes	LDA	LDA on K covariances	epochs
Cov. Align.	yes	LDA	align + LDA on K	epochs

class weighting but differ in test-time adaptation (a logistic-regression probe vs. gradient steps); Table I summarizes where every method sits on data access, class weighting, adaptation, and input.

E. Implementation details

MAML: inner LR 0.01, 5 inner steps, outer LR 0.001 (Adam, weight decay 10^{-4}). ANIL: same, inner loop on head only. Reptile: 5 inner steps, outer step 0.1. SubjectConditioned: 32-dim subject embedding from the prototype trio, FiLM γ/β from an embedding MLP. Meta-training runs 2000 meta-iterations with a meta-batch of 4 subjects; each episode is balanced with 10 support and 40 query trials (5/20 per class). Test-time adaptation uses 5 inner steps; for the episodic methods (meta- and metric-based) the K-shot score is averaged over 10 independent calibration draws, whereas the single-shot baselines (Supervised, Pretrain, EEGNet, Riemannian) use one balanced draw. The INRIA recordings are sampled at 200 Hz and the coadaptation recordings at 250 Hz, so EEGNet's temporal kernel length ($f_s/2$) is 100 and 125 samples respectively. PCA retains 32 whitened components, fit per fold on the training subjects only. All hyperparameters were fixed *a priori* from the cited source papers and a single pilot subject; none were tuned on the held-out (test) subjects, so the LOSO estimates are leakage-free. Code, trained per-fold checkpoints, an `analysis.py` that regenerates every table and test, and the exact preprocessing/episode configuration (including the per-fold retained PCA variance) are released (anonymized for review): https://anonymous.4open.science/r/errp-meta.

F. Protocol and statistics

We evaluate LOSO across all subjects, with three random seeds (42, 123, 456) and $K \in \{5, 10, 20\}$. We report balanced accuracy and AUROC, both robust to the class imbalance, as mean $\pm$ std over the three seeds. We do not tabulate F1: binary F1 depends on which class is labelled positive (the two corpora use opposite codings) and rewards majority-class prediction—e.g. the majority-predicting Pretrain-ZeroShot attains a high binary F1 despite near-chance balanced accuracy—so it is misleading here; the error class is the minority in both corpora (prior ∼0.32 INRIA, ∼0.20 coadaptation). The unit of analysis for inference is the *subject*: under LOSO the three seeds are repeated measurements of the same 16 held-out subjects, not independent samples, so we first average each subject's score across seeds and run paired Wilcoxon signed-rank tests at $N = 16$ (exact null distribution). We apply Benjamini–Hochberg false-discovery-rate (FDR) correction across the full set of 69 comparisons per dataset (23 method pairs $\times$ 3 values of K) and report the rank-biserial effect size $r = Z/\sqrt{N} \in [-1, 1]$. With $N = 16$ the smallest attainable two-sided exact p is $2/2^{16} \approx 3.1 \times 10^{-5}$; we report values at this floor as such rather than extrapolating a normal tail. Because r and the p-value both saturate once all 16 subjects move the same way, they convey direction but not magnitude; for the headline comparisons we therefore also report the median paired difference with a 95% bootstrap confidence interval over subjects (10,000 resamples). The $\pm$ in the tables is the spread across the three seeds (run-to-run stability), *not* a subject-level confidence interval; the bootstrap intervals quantify the latter.

TABLE II
PRIMARY DATASET (INRIA NER 2015): BALANCED ACCURACY AT $K \in \{5, 10, 20\}$ AND AUROC AT K=5, MEAN $\pm$ STD OVER THREE SEEDS, LOSO (N=16 SUBJECTS). † SIGNIFICANTLY BETTER THAN SUPERVISED AT K=5 (PAIRED WILCOXON ON PER-SUBJECT MEANS, BH-FDR $p < 0.05$). ‡ NOT SIGNIFICANTLY DIFFERENT FROM EEGNET AT K=5. EEGNET OVERTAKES THE META-LEARNERS AT K=10–20.

Method	K=5	K=10	K=20	AUROC
EEGNet†	$0.600 \pm .009$	$0.628 \pm .004$	$\mathbf{0.645} \pm .011$	0.640
Reptile†‡	$0.594 \pm .002$	$0.600 \pm .003$	$0.607 \pm .003$	0.637
Subj.-Cond.†‡	$0.584 \pm .007$	$0.584 \pm .009$	$0.587 \pm .013$	0.617
Full-MAML†‡	$0.584 \pm .009$	$0.590 \pm .011$	$0.590 \pm .008$	0.625
MAML-ANIL†‡	$0.581 \pm .013$	$0.580 \pm .015$	$0.585 \pm .015$	0.614
Prototypical	$0.536 \pm .005$	$0.553 \pm .009$	$0.569 \pm .002$	0.548
Matching	$0.535 \pm .002$	$0.527 \pm .003$	$0.542 \pm .022$	0.546
Cov. Align.	$0.526 \pm .026$	$0.546 \pm .025$	$0.562 \pm .020$	0.524
Riemannian	$0.517 \pm .015$	$0.561 \pm .026$	$0.541 \pm .040$	0.542
Pretrain-ZeroShot	$0.563 \pm .007$	$0.563 \pm .007$	$0.563 \pm .007$	0.601
Pretrain-FT	$0.537 \pm .014$	$0.542 \pm .012$	$0.544 \pm .018$	0.563
Supervised	$0.527 \pm .034$	$0.563 \pm .015$	$0.577 \pm .004$	0.538

IV. RESULTS

A. Primary dataset

At K=5, the gradient-based meta-learners, the FiLM encoder, and EEGNet all significantly outperform the from-scratch supervised baseline (Table II; BH-FDR $p < 0.05$, $r = 0.82$–1.0, $\Delta = 5$–7 pp). No meta-learner significantly differs from EEGNet at K=5 ($p = 0.49$–0.94): at this shot count they are effectively equivalent. The metric-based methods (Prototypical, Matching) and the covariance-based baselines do not significantly beat the supervised baseline at any K, consistent with their need for more samples to stabilize covariance estimates or learned embeddings. As K

TABLE III
SECOND-DATASET REPLICATION (ERRP-COADAPTATION): BALANCED ACCURACY AT $K \in \{5, 10, 20\}$ AND AUROC AT K=5, MEAN ± STD OVER THREE SEEDS, LOSO (N=16 SUBJECTS). † SIGNIFICANTLY BETTER THAN SUPERVISED AT K=5 (BH-FDR $p < 0.05$). AT K=5 THE GRADIENT META-LEARNERS, THE FILM ENCODER, AND PROTOTYPICAL ALSO BEAT EEGNET (BH-FDR $p < 0.005$); AT K=10–20 THEY DO NOT DIFFER SIGNIFICANTLY FROM EEGNET, WHICH IS THE TOP METHOD AT K=20. BOLD = BEST AT THAT K.

Method	K=5	K=10	K=20	AUROC
Reptile†	**0.720** ± .003	0.724 ± .005	0.731 ± .006	0.799
Subj.-Cond.†	0.721 ± .006	0.724 ± .004	0.723 ± .008	0.800
MAML-ANIL†	0.717 ± .001	0.720 ± .001	0.722 ± .001	0.799
Full-MAML†	0.716 ± .003	0.720 ± .001	0.724 ± .001	**0.801**
Prototypical†	0.681 ± .007	**0.727** ± .002	0.732 ± .008	0.744
EEGNet†	0.654 ± .004	0.713 ± .009	**0.749** ± .003	0.754
Matching†	0.650 ± .009	0.674 ± .006	0.678 ± .013	0.692
Pretrain-ZeroShot†	0.700 ± .001	0.700 ± .001	0.700 ± .001	0.767
Pretrain-FT	0.590 ± .021	0.643 ± .013	0.653 ± .014	0.649
Riemannian	0.514 ± .007	0.566 ± .014	0.584 ± .007	0.501
Cov. Align.	0.521 ± .030	0.566 ± .009	0.564 ± .009	0.534
Supervised	0.563 ± .013	0.581 ± .018	0.614 ± .011	0.593

grows, EEGNet pulls ahead: at K=20 it significantly exceeds Full-MAML, MAML-ANIL, and the FiLM encoder (0.645 vs. 0.585–0.590; BH-FDR $p < 0.02$, $r = 0.72$–0.79) and is numerically ahead of Reptile (0.645 vs. 0.607). Meta-learning's advantage on this dataset is thus specific to the extreme low-data regime, where it merely matches—rather than beats—a source-pretrained CNN. The strongest gain over Supervised at K=5 is Reptile's (median paired difference $+0.051$, 95% bootstrap CI $[+0.038, +0.086]$). AUROC tracks balanced accuracy across K (e.g. EEGNet $0.640/0.679/0.693$ and Reptile $0.637/0.642/0.648$ at K=5/10/20); the per-K AUROC table is released with the code.

B. Replication on a second dataset

The more demanding test is whether the finding reproduces on a structurally different paradigm (run as a separate LOSO experiment, not trained across corpora). It does, and most clearly at K=5. On the coadaptation data—a different task, montage, and stimulus timing—all meta-learners beat the supervised baseline by roughly 15–16 points ($p = 3.1 \times 10^{-5}$, the exact floor at N=16, $r = 1.0$; Reptile median $+0.159$, 95% CI $[+0.123, +0.187]$). Here several families also beat the source-pretrained EEGNet at K=5—not only the gradient meta-learners (Reptile median $+0.065$, CI $[+0.035, +0.086]$) but also the FiLM encoder ($+0.073$) and Prototypical networks ($+0.021$, $p < 0.005$); Matching does not. This EEGNet gap is confined to K=5: by K=10 the meta-learners and EEGNet no longer differ significantly ($p = 0.40$–0.63), and at K=20 EEGNet is the strongest method (0.749). The across-seed spread of the meta-learners is only 0.001–0.008 balanced accuracy, so the K=5 result is not seed-dependent. The covariance-based baselines again fail to beat supervised—in fact they fall significantly below it here—reproducing the same negative result on both datasets.

C. Meta-learning objective vs. data access

Because the meta-learners see every source subject while the from-scratch supervised baseline does not, part of their edge could reflect data access rather than the meta-learning objective. The two fair-transfer baselines control for this directly: both pretrain on the same $N-1$ source subjects *and* use the same inverse-frequency class weighting as the meta-learners. They differ from the meta-learners in two ways—no episodic meta-objective, and a logistic-regression probe instead of gradient adaptation at test time—so the gap isolates the meta-learning *approach* jointly, not the objective in isolation. The meta-learners significantly outperform *both* on *both* datasets at every K. On the primary data at K=5, Reptile exceeds Pretrain-FT by 5.7 points (BH-FDR $p < 0.01$, $r = 0.77$) and Pretrain-ZeroShot by 3.1 points ($p < 0.01$, $r = 0.80$); on the validation data the gaps widen to roughly 13 points over Pretrain-FT ($p = 3.1 \times 10^{-5}$, $r = 1.0$) and 2 points over the strong zero-shot transfer ($p < 0.01$, $r = 0.73$). Two further observations sharpen the picture. First, naive K-shot fine-tuning is not itself a reliable win: Pretrain-FT does not significantly beat the from-scratch supervised baseline on the primary dataset (and is numerically worse at K=20), and at K=5 it trails its own zero-shot variant—fitting a head on five noisy trials adds more variance than signal. Second, simple zero-shot transfer is a surprisingly strong baseline on the validation paradigm (0.700, above EEGNet at K=5), yet the meta-learners still beat it. Because these baselines match the meta-learners on both data access and class weighting, the residual advantage is attributable to the meta-learning approach (episodic training plus gradient adaptation) rather than to source-subject exposure or imbalance handling; disentangling the objective from the adaptation rule—e.g. giving the transfer baselines the same gradient adaptation—is left to future work.

D. Per-subject analysis

A group mean can hide a method that helps a few subjects and hurts the rest, so Fig. 1 breaks the K=5 result down subject by subject, plotting the best meta-learner (Reptile) against the supervised baseline. Almost every point sits above the diagonal: 16 of 16 subjects improve on the validation data and 15 of 16 on the primary data. The benefit is broad, not the work of a handful of outliers.

V. ABLATIONS

Which design choices actually carry the result? Fig. 2 isolates three of them on the primary dataset. **Class-weighted vs. uniform episodic loss.** This is the most impactful choice. Weighting the loss by inverse class frequency lifts Full-MAML balanced accuracy by 2–3 points and prevents the model from collapsing onto the majority (correct) class—a real risk given the 75/25 split. **FiLM vs. no-FiLM.** FiLM conditioning gives a small but significant gain at K=5 ($+1.5$ pp; paired Wilcoxon, N=16, p=0.009, r=0.65) that disappears by K=10 ($p > 0.27$; Fig. 2b). It is thus a minor, low-shot-specific contributor rather than a primary driver. **Feature**

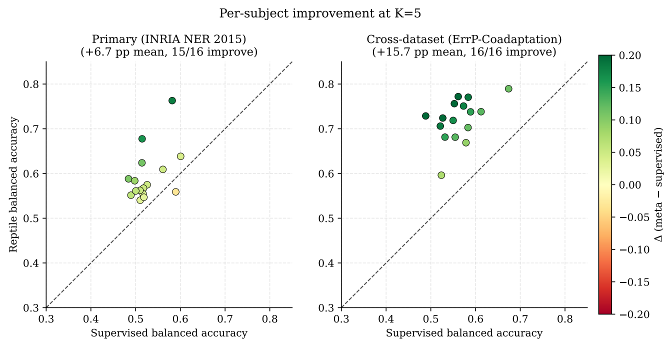

Fig. 1. Per-subject balanced accuracy at K=5: Reptile vs. supervised. Points above the diagonal are subjects where meta-learning helps; color encodes the gain.

dimensionality (32 vs. 128 PCA components). Quadrupling the PCA rank of the flattened waveform from 32 to 128 components did not improve the meta-learners, indicating that 32 components already capture the discriminative ErrP morphology and that the aggressive compression is not the bottleneck.

VI. Discussion

Two factors drive the few-shot gains: class-weighted episodic loss, which prevents majority-class collapse under the 75/25 ErrP imbalance, and cross-subject pre-training that produces a backbone from which only the classifier head needs personalizing. This is consistent with the analysis of Raghu et al. [11]: MAML-ANIL (head-only adaptation) matches Full-MAML throughout our experiments (0.581 vs. 0.584 at K=5 primary; 0.717 vs. 0.716 at K=5 validation), indicating that the shared backbone is the key mechanism and full inner-loop adaptation over the feature extractor adds little.

The results should be interpreted with their scope in mind. The meta-learners' advantage over the strongest neural reference (EEGNet) is confined to the five-trial regime, and is clearest on the second dataset: there the gradient meta-learners, the FiLM encoder, and Prototypical networks all beat EEGNet at K=5, but the gap closes by K=10 and reverses by K=20 (EEGNet 0.749), and on the primary dataset they only match EEGNet at K=5 before EEGNet pulls ahead. We emphasize that this is *replication* on a second paradigm, not trained transfer across corpora. Meta-learning's practical niche is therefore narrow—the extreme low-data regime. The covariance-based methods consistently fail at small K on both datasets, consistent with their need for well-conditioned covariance estimates.

Access to source-subject data does not by itself explain the gains. The meta-learners outperform two transfer baselines (Pretrain-FT, Pretrain-ZeroShot) that pretrain on the same source subjects with the same class-weighted loss, on both datasets and at every shot count. These controls are matched on data access and class weighting (the latter being the dominant ablation factor) but still differ in test-time adaptation, so we attribute the residual gap to the meta-learning approach—episodic training together with gradient adaptation—rather than to data volume or imbalance handling alone.

Statistical vs. practical significance. A significant improvement is not the same as a usable decoder. On the primary dataset the best methods reach only ∼0.60 balanced accuracy at K=5—well above the 0.527 from-scratch baseline in relative terms, but far from a deployable single-trial ErrP detector, which for closed-loop correction would need both high error-class recall and low false-alarm rates at an operating threshold. The cross-dataset numbers (∼0.72) are more encouraging but still short of deployment. We therefore frame the contribution as evidence that meta-learning improves low-data personalization, not as a claim of deployment readiness; cost-sensitive thresholding and online evaluation are needed before practical use.

Limitations. Several caveats bound these claims. (i) *Input is not matched across method families* (Table I): the meta-learners use 32-component PCA features while EEGNet and the Riemannian methods use the raw signal, so "meta-learning vs. EEGNet" partly compares input formats; an input-matched comparison (EEGNet on PCA features, or a meta-learner on EEGNet's raw input) is needed to fully separate the two. (ii) *EEGNet is used frozen* with a logistic-regression probe; an end-to-end fine-tuned EEGNet—plausibly stronger, especially at $K{\geq}10$ where EEGNet already leads—is not included. (iii) The K-shot calibration set is *class-balanced*, so the headline "five trials" is five balanced trials; obtaining $K/2$ error trials naturally requires more than K raw trials at a 20–30% error rate, and a deployed user gets a single draw rather than the 10-draw average used for the episodic methods. (iv) All experiments are offline simulations on two datasets and one backbone family (3-layer MLP / EEGNet) across three seeds; we report balanced accuracy and AUROC but not an operating-point precision/recall (error-recall vs. false-alarm) analysis, which real-time correction would require. Additional datasets, an input-matched and fine-tuned EEGNet, larger backbones, more seeds, and an online closed-loop evaluation would each strengthen the benchmark.

VII. Conclusion

On two independent ErrP paradigms, gradient-based meta-learning, a FiLM-conditioned encoder, and a source-pretrained EEGNet all significantly improve over a from-scratch supervised baseline in the five-trial calibration regime, and the finding *replicates* on a second, structurally different paradigm. There, at K=5, the gradient meta-learners, the FiLM encoder, and Prototypical networks all outperform EEGNet, with seed-robust effects and broad per-subject coverage (15–16 of 16 subjects improve); this edge over EEGNet is, however, specific to K=5 and does not hold at K=10–20. The main design lesson is that class-weighted episodic loss matters more than architecture, and head-only adaptation (ANIL) matches full inner-loop MAML throughout. Against transfer baselines matched on source-data access and class weighting, the meta-learners still win, locating the gain in the meta-learning approach rather than data access. The absolute accuracies

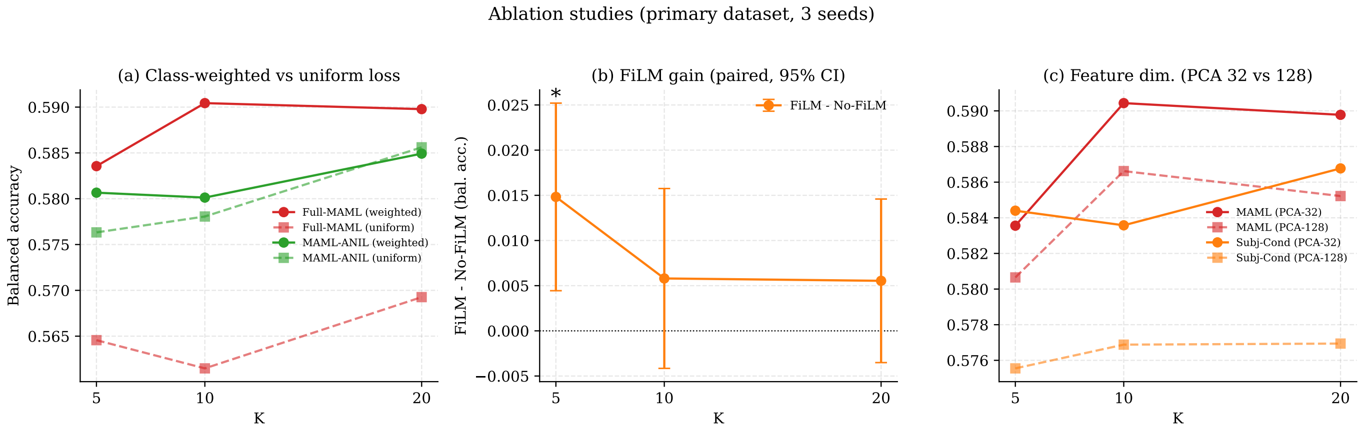

Fig. 2. Ablations (primary dataset). (a) class-weighted vs. uniform loss; (b) the FiLM gain plotted as the *paired* per-subject difference (FiLM − No-FiLM) with a 95% confidence interval across subjects—matching the within-subject test—which excludes zero only at $K{=}5$ ($*$); (c) 32 vs. 128 PCA components.

($\sim$0.60 primary, $\sim$0.72 second dataset at $K{=}5$) are not yet deployment-grade; future work should add an input-matched and end-to-end fine-tuned EEGNet, extend to additional ErrP datasets, and validate in a closed-loop online setting.

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